fragilities and trade-offs (Canby et al., 2025): Zhang & Nanda (2024) showed that such approaches are sensitive to experimental design. Then, McGrath et al. (2023) discovered a "hydra effect," where ablating components identified as causally important fail to change behavior due to redundant causal pathways. This phenomenon, known as **overdetermination**, occurs when multiple redundant, independently sufficient causal pathways exist (Schaffer, 2003; Sider, 2003; Dyrkolbotn, 2017). Rather than isolating simple mechanisms, interventions tend to reveal overdetermined causal structures.

Mechanistic Interpretability. Causal approaches culminate in *mechanistic interpretability* (MI), which aims to reverse-engineer networks into human-interpretable algorithms (Olah et al., 2020). One family of approaches (*where-then-what*) first identifies circuits carrying information from inputs to outputs and then interprets their components (Dunefsky et al., 2024; Davies & Khakzar, 2024; Conmy et al., 2023). The second (*what-then-where*) instead starts from high-level candidate algorithms and searches for causally aligned neural subspaces, using *causal abstraction* metrics (Geiger et al., 2022a;b; Beckers & Halpern, 2019). Despite promising demonstrations, both categories have the typical issues (Sharkey et al., 2025). Subspace patching can produce *interpretability illusions* by activating alternate pathways (Makelov et al., 2023), also a problem of overdetermination. Circuit explanations often fail to generalize (Wang et al., 2022; Li et al., 2025b) and are sensitive to minor experimental choices (Méloux et al., 2025). Exhaustive studies on toy models reveal multiple incompatible explanations for both strategies, even for random networks (Méloux et al., 2025). Finally, Sutter et al. (2025) proved that, in general, existing causal abstraction methods can produce explanations for random networks.

Natural Language Explanations. Generating natural language rationales has become a popular interpretability approach (Marasovic et al., 2022; Wiegreffe et al., 2022). However, Ajwani et al. (2024) showed that LLM-generated explanations can be systematically unfaithful, confidently providing plausible-sounding justifications for predictions made for entirely different reasons. Moreover, chain-of-thought (self-)explanations are typically unfaithful to the model's computation (Lanham et al., 2023; Arcuschin et al., 2025; Turpin et al., 2023). There exist infinite plausible stories that can rationalize any behavior post hoc. Thus, natural language explanations are particularly susceptible to confabulation and, thus, to false positives.

3 The Deeper Statistical Issue

The problems documented in Section 2 point to a broad statistical fragility. Here, we identify the common structure underlying these failures: the non-identifiability of interpretability queries.

Behavior-based approaches that study input–output relationships (e.g., feature attributions, behavioral testing) are fundamentally limited by **underspecification**: multiple, distinct explanations can equally well account for the same input–output patterns (Jacovi et al., 2021; Rogers et al., 2021; Hagendorff et al., 2023). Similar observations in cognitive science motivated the development of brain imaging as a complement to purely behavioral data, with the goal of measuring neural computation and thereby obtaining objective, measurable, and more generalizable quantities (Kosslyn, 1999; Logothetis, 2008; Churchland & Sejnowski, 1988). AI interpretability has followed a related trajectory moving toward analyzing internal computation (Mueller et al., 2024). However, predictive approaches based on internal states (probing, SAEs) inherit standard machine-learning pathologies such as overfitting and poor generalization (Belinkov, 2022). These failure modes are instances of **underspecification**: many predictive models can fit the training data equally well, leaving it unclear which ones posit generalizable causal mechanisms (Teney et al., 2022; D'Amour et al., 2022).

Causal approaches, introduced in response to the shortcomings of predictive methods, appear at first to provide the scientific rigor needed for generalizable explanations. However, AI systems are large, distributed systems with many interacting components, which gives rise to redundant and context-dependent causal pathways (Frankle & Carbin, 2019). This creates **overdetermination**, where multiple distinct causal mechanisms are each independently sufficient to produce the same behavior (Tononi et al., 1994; Loosemore, 2012; Sarkar, 2022). Then, finding mechanistic stories within complex computational systems can become *too easy*: many different, incompatible explanations can be produced for the same phenomenon (Lindsay & Bau, 2023; Méloux et al., 2025).

Identifiability. These failure modes can be formalized using the concept of identifiability. Informally, identifiability is the property of a statistical inference task stating that the parameters (explanatory variables) of a statistical model can be uniquely recovered from available observations (Casella & Berger, 2024). Identifiability is typically a prerequisite for reliable inference in the natural sciences; without it, inferred explanations remain ambiguous. Therefore, substantial work in statistics, unsupervised learning, and causal inference has focused on characterizing identifiability conditions and designing identifiable tasks (Casella & Berger, 2024; Allman et al., 2009; Locatello et al., 2019; Khemakhem et al., 2020; Shpitser & Pearl, 2008).

For interpretability, both underspecification and overdetermination produce non-identifiability, explaining most of the statistical fragilities: (i) **Poor generalization:** when multiple explanations fit the observed data equally well, their explanatory claims can diverge arbitrarily on unseen data. Selecting among these explanations, therefore, depends on arbitrary inductive biases that are rarely validated. (ii) **Sensitivity to design choices:** non-identifiability implies a manifold of explanations that achieve a *good fit*. Different algorithmic choices (datasets, optimization procedures, hyperparameters) traverse this manifold differently, and thus produce different explanations. (iii) **False discovery:** when explanations are non-identifiable, the probability of recovering a spurious explanation that happens to fit the data increases with the size and complexity of the hypothesis space.

Currently, identifiability is just a conceptual analogy, because interpretability has not yet been formalized as an explicit statistical inference task. Making this formal connection and casting interpretability queries as well-specified statistical estimation problems is a necessary first step toward developing methods whose limitations and assumptions can be explicitly characterized.

4 The Statistical–Causal Inference Perspective

A straightforward way to address dead-salmon artifacts across interpretability methods is to compare findings on a trained target network against a randomized alternative: the same architecture with randomized weights analyzed by the same method. This leads to a principled hypothesis test against a null hypothesis of randomized computation, an idea foreshadowed in early work on probing (Conneau et al., 2018; Hewitt & Manning, 2019; Ravichander et al., 2021) and circuit discovery (Shi et al., 2024). We formalize such a test in Appendix A and show that, for probing, it eliminates some false discoveries and substantially reduces effect sizes in standard analyses.

While effective, directly correcting dead-salmon artifacts is a very low bar for interpretability. The goal is to address the deeper statistical issues that give rise to these failures in the first place. Nevertheless, hypothesis testing against computationally meaningful null alternatives naturally motivates a broader statistical–causal reframing of interpretability.

Here, we sketch one such formalization, viewing interpretability as a problem of *statistical–causal inference*. In this view, explanations are *surrogate models* constructed to answer distributions of causal queries about a computational system. An explanation is useful insofar as it supports prediction and manipulation, generalizes under intervention, and remains robust to noise. This perspective aligns with a growing pragmatist approach to interpretability (Páez, 2019).

4.1 Background: Statistical–Causal Inference

Statistical inference provides the rigorous framework through which empirical observations become scientific knowledge (Cox, 2006; Lehmann & Casella, 1998). We argue that interpretability, like every empirical science, must be grounded in these principles. We provide here a brief overview.

Statistical Models and Identifiability. A *statistical model* is a family of probability laws $\{\mathbb{P}_\theta^{\mathbf{V}} : \theta \in \Theta\}$ on a sample space $\mathcal{V}$, indexed by parameters $\theta \in \Theta$. Here, $\mathbf{V}$ denotes observed data. Intuitively, we assume data arises from some process indexed by unknown parameters θ, and the goal is to recover θ from observations. Sound inference requires *identifiability*: distinct parameters must induce distinct distributions over observables. Formally, a model is identifiable if $\theta \neq \theta' \implies \mathbb{P}_\theta^{\mathbf{V}} \neq \mathbb{P}_{\theta'}^{\mathbf{V}}$. Without identifiability, hypotheses cannot be distinguished from data, rendering inference ill-posed.

Estimators and Uncertainty Quantification. Given finite observations $\mathcal{D}_n = \{\mathbf{v}^{(i)}\}_{i=1}^n$, an *estimator* T produces an estimate $\hat{\theta} := T(\mathcal{D}_n)$ of unknown parameters θ. Its quality can be assessed through various statistical properties: (i) **Bias**: Does it recover the correct parameter on average? (ii) **Variance**: How much does the estimate vary across datasets? (iii) **Consistency**: Does it converge to the correct parameter as $n \to \infty$? Beyond point estimates, **confidence sets** provide uncertainty quantification under finite sampling.

Causal Inference. Many scientific questions go beyond prediction, seeking explanations of *how* variables influence one another. This requires enriching statistical models with a causal structure (Pearl, 2009). Let $\mathbf{V} = \{V_1, \ldots, V_d\}$ denote *endogenous variables*, quantities computed within the system. A directed graph $\mathcal{G}$ over nodes $\mathbf{V}$ encodes direct causal relationships: an edge $V_i \to V_j$ indicates that V_i directly causes V_j. A *structural causal model* (SCM) is the tuple $\mathfrak{C} = (\mathbf{V}, \mathbf{U}, \mathbf{f}, P_{\mathbf{U}})$, where:

- $\mathbf{U}$ collects *exogenous* (external) inputs representing unobserved causes or environmental randomness, $P_{\mathbf{U}}$ is their joint distribution
- $\mathbf{f} = \{f_1, \ldots, f_d\}$ are *structural assignments*, functions that deterministically compute each variable from its causes:

$$V_i = f_i(\mathbf{PA}_i, U_i), \qquad i = 1, \ldots, d, \tag{1}$$

where $\mathbf{PA}_i \subseteq \mathbf{V}$ denotes the parents of V_i in $\mathcal{G}$, and $U_i \in \mathbf{U}$ is its exogenous input.

SCMs enable reasoning about interventions and counterfactuals. A *hard intervention* $\mathrm{do}(\mathbf{V}_I = \mathbf{v}_I)$ on a subset $\mathbf{V}_I \subseteq \mathbf{V}$ replaces the structural assignments for variables in $\mathbf{V}_I$ with constants $\mathbf{v}_I$, overriding their causal mechanisms. The intervened model $\mathfrak{C}; \mathrm{do}(\mathbf{V}_I = \mathbf{v}_I)$ induces an *interventional distribution* $P_{\mathbf{V}}^{\mathfrak{C};\mathrm{do}(\mathbf{V}_I = \mathbf{v}_I)}$, which captures how the system behaves under this external manipulation.

A *causal query* $q(\mathfrak{C})$ is any well-defined question about the SCM, such as "What is the marginal distribution of V_i?" or "What is the average effect of setting $V_i = v$ on outcome V_m?" Thus, a causal query is any measurable functional of the SCM, possibly involving conditioning or intervention. Central to causal inference is *query identifiability*: whether $q(\mathfrak{C})$ can be uniquely determined from available observational or interventional data.

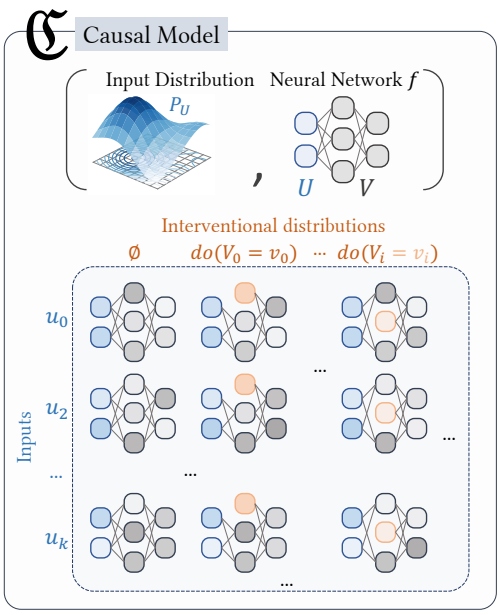


Figure 2: The tuple of a target behavior $P_{\mathbf{U}}$ and the computational system with its internal components form an SCM.

4.2 Neural Networks as Structural Causal Models

Returning to modern AI interpretability, we first state a standard framing of computational systems as SCMs. Let f be a computational system, typically a neural network, with internal computational elements $\mathbf{V}$ and input distribution $P_{\mathbf{U}}$. The input distribution $P_{\mathbf{U}}$ represents the *behavior of interest* that we aim to explain. For instance, $P_{\mathbf{U}}$ might represent arithmetic prompts to a language model, images from a particular domain, or factual questions about a specific topic.

The tuple $(f, P_{\mathbf{U}})$ naturally defines an SCM $\mathfrak{C} = (\mathcal{G},\ \mathbf{V},\ \mathbf{U},\ \mathbf{f},\ P_{\mathbf{U}})$, where:

- **Endogenous variables** $\mathbf{V}$ are the network's computational variables (e.g., hidden states, attention patterns, outputs).